Homework Grading Report

Student:	Student_1
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	**10.0**	All tasks completed
Part 2: Data Cleaning	15	**7.5**	Missing: colSums(), na.omit()
Part 3: Basic Data Transformation	15	**15.0**	Missing: %>%()
Part 4: Advanced Transformation	15	**15.0**	All tasks completed
Part 5: Data Reshaping	15	**15.0**	All tasks completed
Part 6: Combining Datasets	15	**15.0**	All tasks completed
Part 7: String Manipulation & Date/Time	20	**20.0**	All tasks completed
Part 8: Advanced Wrangling	15	**15.0**	All tasks completed
Part 9: Reflection Questions	5	**5.0**	All tasks completed
TOTAL	**125**	**119.3**	**95%**

Detailed Feedback

What You Did Well

- Your approach to calculating outlier thresholds using the IQR method in Part 2.3 is correct and well-implemented with clear variable naming.

- Your pipe chaining in Part 3.3 for sorting and selecting top sales demonstrates good understanding of dplyr operations and the pipe operator.

- Your reshaping operations in Parts 5.2 and 5.3 using pivot_wider and pivot_longer functions show solid grasp of data reshaping techniques.

- Your business KPI calculation in Part 8.2 correctly computes total revenue, transaction count, average value, and percentage with proper summarization syntax.

- Your string cleaning approach in Part 7.1 properly applies str_trim and str_to_title functions to clean text fields effectively.

- You correctly identified and removed rows with missing values, which prevents distortion of summary statistics and ensures that revenue calculations are based on complete transactions.

What to Fix

1. **Task 2.1: Check for Missing Values:** Your code attempts to create missing_summary but fails because sales_data is not defined in your environment. You need to load the dataset first. The correct approach: missing_summary <- sapply(sales_data, function(x) sum(is.na(x))) or better yet: missing_summary <- colSums(is.na(sales_data)).

2. **Task 2.2: Handle Missing Values:** While your code executes without error, it appears you're working with a dataset that has no missing values since original rows and cleaned rows both equal 300. This suggests either the data was already clean or the sales_data wasn't properly loaded.

3. **Task 7.2: Parse Dates and Extract Components:** Your code fails to parse dates due to format mismatch. The warning indicates all formats failed to parse. You need to adjust the date parsing function to match your actual date format. The correct approach: try using ymd_hms() or specify the exact format like mdy('%m/%d/%Y') depending on your data format.

4. **Task 7.1: Clean Text Data:** Your code works syntactically but may not be fully effective. You're applying str_to_title to the entire Region field which might not be desired. For example, 'Latin America' becomes 'Latin America' rather than 'Latin america'. Consider using str_to_title(str_trim(Region)) for better results.

5. **Task 8.2: Calculate Business KPIs:** Your code correctly calculates all KPIs but you're missing the high_value column in your dataset. You need to ensure this column exists before calculating the percentage. The correct approach: make sure high_value column exists in sales_enhanced before running the summarize operation.

6. To strengthen your reflection on data cleaning, include specific metrics (e.g., number of rows removed, change in total revenue) and explain why those changes matter for decision-making; the solution example reports that 12 rows were dropped, reducing revenue by $45,000.

7. When discussing grouped analysis, cite actual figures from your summaries (e.g., "North region generated $1.2M, which is 20% higher than the national average") and explain how those numbers guide resource allocation; this level of detail demonstrates business impact.

8. Your explanation of wide vs. long formats would benefit from a concrete scenario, such as using wide format for a quarterly sales matrix presented to executives and long format for a time-series line chart in a marketing dashboard; the solution provides these exact use cases.

Reflection & Critical Thinking

You answered Question 9.1 with a generic statement about data cleaning improving accuracy, but you did not explain how the specific handling of NAs and outliers changed your results (e.g., rows removed, revenue totals adjusted). Providing concrete numbers would demonstrate the tangible impact of cleaning on business insight.

In Question 9.2 you noted that regional and category summaries reveal trends, yet you did not describe any specific patterns you observed (such as which region generated the most revenue or which product category lagged). Detailing those findings would show you can translate grouped statistics into actionable business decisions.

Your response to Question 9.3 correctly identifies that wide format aids side-by-side comparison and long format supports charting, but you did not give a concrete business scenario (e.g., quarterly sales dashboard vs. time-series forecasting). Adding a realistic example would illustrate your understanding of when each shape is most useful.

For Question 9.4 you correctly distinguished left_join from inner_join, but you did not discuss potential pitfalls (e.g., unintentionally dropping customers with no orders) or when you might prefer a right_join or full_join. Explaining these nuances would deepen your grasp of join strategy in real-world data pipelines.

In Question 9.5 you selected joining datasets as the most valuable skill, yet you did not connect this choice to other lessons (e.g., how joins enable the KPI calculations you performed). Linking the skill to a broader workflow would demonstrate integrated thinking across the course.

Instructor Assessment

You have completed all required coding tasks and your code runs without errors, which shows solid technical competence. However, your reflection responses are overly brief and lack the quantitative detail needed to demonstrate true business insight. Strengthen your answers by referencing specific numbers from your analysis and explaining how those figures influence strategic decisions. Completing this step will significantly improve both your analytical depth and your overall grade.

Key Takeaway

Revise each reflection answer to include quantitative results from your code and a clear link to business implications.

Practice articulating the business rationale behind each data-wrangling step, using concrete examples from the dataset.

Add brief code snippets or output excerpts to your reflections to demonstrate evidence-based reasoning.

Review the solution notebook to see how detailed insights (e.g., specific revenue figures, percentage changes) are presented and emulate that depth in your own answers.
